

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

First Semester 2024-25

CE F213: Surveying Lab Quiz – 2: Solutions

Instructor-In-Charge: Prof. S.N. Patel

Name: _____

Date: 27.11.2024

ID No: _____

Max Marks: 40

Instructions:

Duration: 50 minutes

1. Each question carries **ONE** mark 2. No negative marking 3. Calculators are allowed

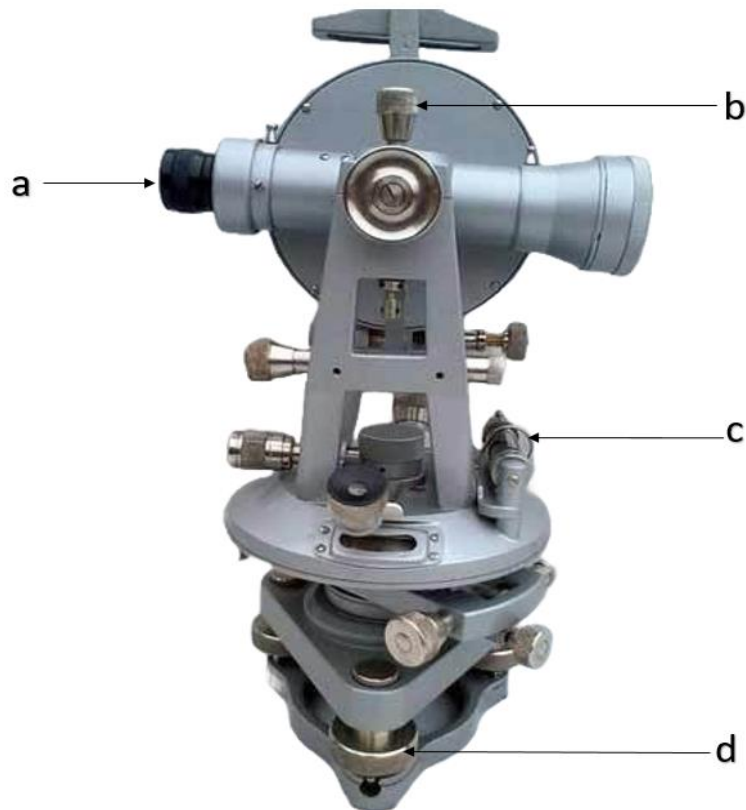
1. Which of the following is NOT the temporary adjustment required to make a theodolite ready for taking observation at a station?
 - a. Centring
 - b. Setting up
 - c. Levelling up
 - d. **Calibration of theodolite**
2. Assertion (A): To determine the vertical angle using a theodolite, the average of vertical readings from face left and face right positions is calculated.
Reasoning (R): The process involves setting up the instrument, rotating the telescope horizontally to bisect the object, clamping the plates, and using the vertical tangent screws to precisely bisect the target point.
 - a. Both (A) and (R) are true, and (R) is the correct explanation of (A).
 - b. **Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).**
 - c. (A) is true, but (R) is false.
 - d. (A) is false, but (R) is true.
3. Why slip error occurs in a theodolite?
 - a. **This error will occur if clamp screw is not properly tightened.**
 - b. This error is due to manipulating wrong tangent screw.
 - c. This error will come when inaccurate bi-section of points observed.
 - d. No, there is no such error in any measuring instrument.
4. In Vernier theodolite the Vernier's are fitted to read the angles up to 20".
 - a. **True**
 - b. False
5. The data obtained from total station can be used in which among the following software directly?
 - a. Primavera
 - b. Autodesk Revit
 - c. STAAD Pro
 - d. **Arc GIS**
6. The face left position in theodolite is called _____.
 - a. Telescope inverted
 - b. Telescope reversed
 - c. **Telescope normal**
 - d. None of the above
7. Which method is used to improve the accuracy of measured included angles?
 - a. Reiteration
 - b. **Repetition**
 - c. Easting
 - d. Deflection
8. Which of the following indicates the correct set of the combination of total station?
 - a. Theodolite, compass.
 - b. Autolevel, EDM.
 - c. **Electronic theodolite, EDM.**
 - d. EDM, GPS, Tape, Chain.
9. The radius of one-degree curve for 20m arc length is _____.
 - a. 1719 m
 - b. 2000 m
 - c. **1146 m**
 - d. None of the above

10. The point where circular curve starts is **Point of Curvature (P.C)**, and the point where circular curve ends is **Point of Tangency (P.T)**.
11. In offsets from chord produced method, the n^{th} or last offset is calculated by _____.
- $O_n = \frac{C_n}{2R} (C_{n+1} + C_n)$
 - $O_n = \frac{C_n}{2R} (C_{n-1} + C_n)$
 - $O_n = \frac{C_n}{2R} (C_{n-1} - C_n)$
 - $O_n = \frac{C_{n-1}}{2R} (C_{n-1} + C_n)$
12. The internal angle formed at the intersection of forward tangent and backward tangent in simple circular curve is _____.
- Angle of intersection**
 - Deflection angle
 - Point of intersection
 - All of the above
13. The best method for setting out a long curve by linear method and is usually employed for highway curves when a theodolite is not available _____.
- Rankine method
 - Offsets from long chord
 - By offsets from chord produced**
 - All of the above
14. The sum of deflection angle and angle of intersection in simple circular curve (in degrees) is **180°**.
15. In simple curve, let R be the radius of curve and Δ be the deflection angle, then the length of backward tangent is calculated by _____.
- $R \tan (\Delta)$
 - $R \tan (\Delta/4)$
 - $R^2 \tan (\Delta/2)$
 - $R \tan (\Delta/2)$**
16. The chainage at point of intersection is 545 m, radius of curve is 88 m and deflection angle is 100° . The chainage at point of tangency is _____.
- 440.126 m
 - 698.588 m
 - 593.715 m**
 - None
17. What is the length of curve, if the radius of curve is 125 m and deflection angle is 135° ?
- 294.524 m**
 - 198.174 m
 - 289.656 m
 - 104.568 m
18. Which of the following is not a linear method of setting out of simple curve?
- Perpendicular offset method
 - Rankine method**
 - Radial offset method
 - By offsets from chord produced method
19. The operation of revolving the telescope through 180° in a vertical plane about its horizontal axis is **Transiting**.
20. _____ is the central angle subtended by one chain length of chord.
- Intersection angle
 - Deflection angle
 - Degree of curve**
 - None
21. In Rankine's method of deflection angle, the deflection angle is equal to _____ of the angle subtended by the arc at the center.
- 1/4
 - Equal
 - 1/2**
 - None
22. The process of turning the telescope in horizontal plane _____.
- Swinging**
 - Transiting
 - Changing face
 - None

23. The possible obstructions in laying out simple curves are _____.
 a. Obstacles to chaining
 b. Obstruction to vision
 c. The point of intersection is in accessible
 d. **All the above**
24. The vertical circle of Vernier theodolite is graduated from 0^0 to 90^0 .
 a. **True**
 b. False
25. The permanent adjustment of theodolite includes,
 i. Plate level test ii. Centring iii. Bubble tube adjustment
 a. (i) is correct
 b. (i) and (ii) are correct
 c. **(i) and (iii) are correct**
 d. All are correct
26. The horizontal circle of a Vernier theodolite has graduations provided in the quadrantal bearing system.
 a. True
 b. **False**
27. The plane table can be used to locate the points more accurately than by a theodolite.
 a. True
 b. **False**
28. The instrument used to sight an object in plane table survey _____.
 a. **Alidade**
 b. Plumb bob
 c. Trough compass
 d. Spirit level
29. Plane table works on the principle of _____.
 a. **Parallelism**
 b. Intersection
 c. Radiation
 d. All the above
30. Which of the following is not the error due to manipulation and sighting in plane table survey?
 a. Inaccurate Centring
 b. Defective sighting
 c. Defective orientation
 d. **Error of plotting**
31. In which case, the orientation by back sighting is used
 a. When approximate levelling is required
 b. **When it is not possible to set instrument over station mark**
 c. When second point is available for orientation
 d. When speed is important than accuracy
32. The instrument station is setup at a station and rays are drawn to various stations to be plotted. This method of plane table surveying is _____.
 a. Intersection
 b. Traversing
 c. **Radiation**
 d. Resection
33. A building is situated on the far side of a river and is inaccessible. Now the building can be located by _____.
 a. Radiation method
 b. **Intersection method**
 c. Traversing method
 d. Resection method
34. A hairpin shaped brass frame having two arms of equal length used in plane table survey is **Plumbing fork.**
35. During orientation of plane table _____.
 a. Farthest point is sighted
 b. Either (a) or (c)
 c. Nearest point is sighted
 d. **Previous station is sighted**
36. The exact orientation is more important in plane table survey than accurate centring for small-scale maps.
 a. **True**
 b. False

37. Label the parts of theodolite (a, b, c and d) given in Fig. 1.

[4 Marks]



a. Eye piece.

b. Vertical clamp screw.

c. Plate bubble/Bubble tube

d. Levelling screw/Foot screw

Fig.1. Theodolite